

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399431
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Chang; Shih-Ming et al.

Semiconductor processing tool and methods of operation

Abstract

An illumination system includes a plurality of pixels (or spots) that are (or may be) configured in one or more polarization configuration types. The pixels of the illumination system may be configured to promote particular types of polarization (e.g., transverse electric (TE) polarization, transverse magnetic (TM) polarization) to increase pattern contrast while achieving suitable exposure operation throughput. Moreover, the pixels of the pixels of the illumination system may be configured to achieve free-form (arbitrary or freely-configurable) polarization, which permits the polarization of radiation to be tailored to particular exposure operation patterns and other parameters.

Inventors:	Chang; Shih-Ming (Hsinchu, TW), Chen; Minfeng (Hsinchu, TW)
Applicant:	Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)
Family ID:	1000008782973
Assignee:	Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)
Appl. No.:	17/651741
Filed:	February 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230152709 A1	May. 18, 2023

Related U.S. Application Data

us-provisional-application US 63264057 20211115

Publication Classification

Int. Cl.: G03F7/20 (20060101); G02B5/08 (20060101); G02B5/30 (20060101); G03F7/00 (20060101); H01L21/027 (20060101)

U.S. Cl.:

CPC G03F7/70191 (20130101); G02B5/0891 (20130101); G02B5/3041 (20130101); G03F7/70116 (20130101); H01L21/0274 (20130101);

Field of Classification Search

CPC: G03F (7/70191); G03F (7/70116); G03F (7/70075); G03F (7/70566); G02B (5/0891); G02B (5/3041); H01L (21/0274)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2014/0111785	12/2013	Hennerkes	359/845	G02B 26/0816
2018/0224751	12/2017	Voogd	N/A	G03F 7/70508

Primary Examiner: Asfaw; Mesfin T

Attorney, Agent or Firm: Harrity & Harrity, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This Patent Application claims priority to U.S. Provisional Patent Application No. 63/264,057, filed on Nov. 15, 2021, and entitled “SEMICONDUCTOR PROCESSING TOOL AND METHODS OF OPERATION.” The disclosure of the prior Application is considered part of and is incorporated by reference into this Patent Application.

BACKGROUND

(1) As semiconductor device sizes continue to shrink, some lithography technologies suffer from optical restrictions, which lead to resolution issues and reduced lithography performance. In comparison, extreme ultraviolet (EUV) lithography can achieve much smaller semiconductor device sizes and/or feature sizes through the use of reflective optics and radiation wavelengths of approximately 13.5 nanometers or less.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a diagram of an example lithography system described herein.

(3) FIG. 2 is a diagram of an example implementation described herein.

- (4) FIG. 3 is a diagram of an example pixel configuration for an illumination system described herein for use in the lithography system of FIG. 1.
- (5) FIGS. 4-6 are diagrams of example implementations of pixel types described herein.
- (6) FIGS. 7, 8A-8D, and 9 are diagrams of example implementations of configurable pixels described herein.
- (7) FIGS. 10A-10C are diagrams of example pixel configurations for an illumination system described herein for use in the lithography system of FIG. 1.
- (8) FIG. 11 is a diagram of example components of one or more devices of FIG. 1 described herein.
- (9) FIGS. 12-14 are flowcharts of example processes relating to configuring pixels of an illumination system described herein.

DETAILED DESCRIPTION

- (10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (12) As pattern sizes continue to decrease in advanced semiconductor fabrication processes, the ability to transfer a high-contrast image patterns onto a semiconductor substrate in a lithography exposure operation becomes more difficult. Numerical aperture sizes may be increased (to 0.55 or greater, as an example) in advanced semiconductor fabrication processes, which may lead to reduced contrast and reduced interference efficiency. This may result in reduced lithography throughput, reduced pattern quality, reduced semiconductor device yield and performance, and/or an increase in semiconductor defects, among other examples.
- (13) Some implementations described herein provide an illumination system for use in a lithography system (e.g., an EUV lithography system or another type of lithography system) and associated methods of operation. The illumination system includes a plurality of pixels (or spots) that are (or may be) configured in one or more polarization configuration types. In this way, the pixels of the illumination system may be configured to promote particular types of polarization (e.g., transverse electric (TE) polarization, transverse magnetic (TM) polarization) to increase pattern contrast while achieving suitable exposure operation throughput. Moreover, the pixels of the illumination system may be configured to achieve free-form (arbitrary or freely-configurable) polarization, which permits the polarization of radiation to be tailored to particular exposure operation patterns and other parameters.
- (14) FIG. 1 is a diagram of an embodiment of a lithography system 100 described herein. The lithography system 100 includes an EUV lithography system or another type of lithography system that is configured to transfer a pattern to a semiconductor substrate using mirror-based optics. The lithography system 100 may be configured for use in a semiconductor processing environment such

as a semiconductor foundry or a semiconductor fabrication facility.

(15) As shown in FIG. 1, the lithography system **100** includes the radiation source **102** and an exposure tool **104**. The radiation source **102** (e.g., an EUV radiation source or another type of radiation source) is configured to generate radiation **106** such as EUV radiation and/or another type of electromagnetic radiation (e.g., light, EUV light). The exposure tool **104** (e.g., an EUV scanner or another type of exposure tool) is configured to focus the radiation **106** onto a reflective reticle **108** (or a photomask) such that a pattern is transferred from the reticle **108** onto a semiconductor substrate **110** using the radiation **106**.

(16) The radiation source **102** includes a vessel **112** and a collector **114** in the vessel **112**. The collector **114**, includes a curved mirror that is configured to collect the radiation **106** generated by the radiation source **102** and to focus the radiation **106** toward an intermediate focus **116**. The radiation **106** is produced from a plasma that is generated from droplets **118** (e.g., tin (Sn) droplets or another type of droplets) being exposed to a laser beam **120**. The droplets **118** are provided across the front of the collector **114** by a droplet generator (DG) head **122**. The DG head **122** is pressurized to provide a fine and controlled output of the droplets **118**.

(17) A laser source, such as a pulse carbon dioxide (CO₂) laser, generates the laser beam **120**. The laser beam **120** is provided (e.g., by a beam delivery system to a focus lens) such that the laser beam **120** is focused through a window **124** of the collector **114**. The laser beam **120** is focused onto the droplets **118** which generates the plasma. The plasma produces a plasma emission, some of which is the radiation **106**. The laser beam **120** is pulsed at a timing that is synchronized with the flow of the droplets **118** from the DG head **122**.

(18) The exposure tool **104** includes an illuminator **126** and a projection optics box (POB) **128**. The illuminator **126** includes a plurality of reflective mirrors that are configured to focus and/or direct the radiation **106** onto the reticle **108** so as to illuminate the pattern on the reticle **108**. The plurality of mirrors include, for example, a mirror **130a** and a mirror **130b** (referred to herein as an illumination system **130b**). The mirror **130a** includes a field facet mirror (FFM) or another type of mirror that includes a plurality of field facets. The illumination system **130b** includes a pupil facet mirror (PFM) or another type of mirror that also includes a plurality of pupil facets, pixels, or illumination spots. As described herein, the pixels of the illumination system **130b** are arranged (and/or are capable of being configured) to turn on/off, focus, polarize, and/or otherwise tune the radiation **106** from the radiation source **102** to increase or emphasize particular types of radiation components (e.g., transverse electric (TE) polarized radiation, transverse magnetic (TM) polarized radiation). This enables the illumination system **130b** to increase the uniformity or change the intensity distribution of the radiation **106** and increase the contrast of the pattern of the reticle **108** transferred to the semiconductor substrate **110**. Another mirror **132** (e.g., a relay mirror) is included to direct radiation **106** from the illuminator **126** onto the reticle **108**.

(19) The projection optics box **128** includes a plurality of mirrors that are configured to project the radiation **106** onto the semiconductor substrate **110** after the radiation **106** is modified based on the pattern of the reticle **108**. The plurality of reflective mirrors include, for example, mirrors **134a-134f**. In some implementations, the mirrors **134a-134f** are configured to focus or reduce the radiation **106** into an exposure field, which may include one or more die areas on the semiconductor substrate **110**.

(20) The exposure tool **104** includes a substrate stage **136** (e.g., a wafer stage) configured to support the semiconductor substrate **110**. Moreover, the substrate stage **136** is configured to move (or step) the semiconductor substrate **110** through a plurality of exposure fields as the radiation **106** transfers the pattern from the reticle **108** onto the semiconductor substrate **110**. The exposure tool **104** also includes a reticle stage **138** that configured to support and/or secure the reticle **108**. Moreover, the reticle stage **138** is configured to move or slide the reticle through the radiation **106** such that the reticle **108** is scanned by the radiation **106**. In this way, a pattern that is larger than the field or beam of the radiation **106** may be transferred to the semiconductor substrate **110**. A

controller **140** included in the lithography system **100** (e.g., in the exposure tool **104** or another component of the lithography system **100**) is configured to communicate with and/or control actions of various components and/or subsystems of the lithography system **100**, including the radiation source **102** and/or the exposure tool **104**, among other examples. In some implementations, the controller **140** transmits signals to the lithography system **100** and/or the components thereof (e.g., the radiation source **102**, the exposure tool **104**) to cause the lithography system **100** and/or the components thereof (e.g., the radiation source **102**, the exposure tool **104**) to perform an exposure operation.

(21) In an example exposure operation (e.g., an EUV exposure operation), the DG head **122** provides the stream of the droplets **118** across the front of the collector **114**. The laser beam **120** contacts the droplets **118**, which causes a plasma to be generated. The plasma emits or produces the radiation **106** (e.g., EUV light). The radiation **106** is collected by the collector **114** and directed out of the vessel **112** and into the exposure tool **104** toward the mirror **130a** of the illuminator **126**. The mirror **130a** reflects the radiation **106** onto the illumination system **130b**, which reflects the radiation **106** onto the mirror **132** toward the reticle **108**. The radiation **106** is modified by the pattern in the reticle **108**. In other words, the radiation **106** reflects off of the reticle **108** based on the pattern of the reticle **108**. The reflective reticle **108** directs the radiation **106** toward the mirror **134a** in the projection optics box **128**, which reflects the radiation **106** onto the mirror **134b**. The radiation **106** continues to be reflected and reduced in the projection optics box **128** by the mirrors **134c-134f**. The mirror **134f** reflects the radiation **106** onto the semiconductor substrate **110** such that the pattern of the reticle **108** is transferred to the semiconductor substrate **110**. The above-described exposure operation is an example, and the lithography system **100** may operate according to other EUV techniques and radiation paths that include a greater quantity of mirrors, a lesser quantity of mirrors, and/or a different configuration of mirrors.

(22) As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

(23) FIG. **2** is a diagram of an example implementation **200** described herein. The example implementation **200** includes an example operation of the mirror **130a** and the illumination system **130b**. As shown in FIG. **2**, the mirror **130a** includes a plurality of mirror facets **202**. The mirror facets **202** may include rectangular-shaped mirror facets, square-shaped mirror facets, and/or may include another shape of mirror facets. The mirror facets **202** are configured to receive the radiation **106** and split the radiation **106** into individual or separate beams of radiation. In this way, the mirror facets **202** are configured to tune, modify, or adjust the radiation **106**.

(24) As further shown in FIG. **2**, the illumination system **130b** includes a plurality of pixels **204** on a substrate **206**. The pixels **204** include various components described herein, including mirrors, polarizers, and/or actuators, among other examples. The pixels **204** are configured to receive the beams of radiation from the mirror facets **202** and reflect (or redirect) the beams of radiation toward the reticle **108** (or other intervening mirrors). In some implementations, the pixels **204** include approximately circle-shaped structures that are arranged in a grid pattern or another pattern on the substrate **206**. In some implementations, the pixels **204** include microelectromechanical systems (MEMS) that include the mirrors, polarizers, and/or actuators, described herein. In these implementations the MEMS of the pixels **204** (and the mirrors, polarizers, and/or actuators, described herein) may be formed by various MEMS fabrication and/or processing techniques.

(25) In some implementations, the mirror **130a** includes on the order of hundreds of mirror facets **202**. For examples, the mirror **130a** may include 300 or more mirror facets **202** or another quantity of mirror facets **202**. In some implementations, the illumination system **130b** includes on the order of thousands of pixels **204** or more. For examples, the illumination system **130b** may include 1000 or more pixels **204** or another quantity of pixels **204**. In some implementations, a subset of the mirror facets **202** and a subset of the pixels **204** are activated in an exposure operation of the lithography system **100**.

(26) As further shown in FIG. 2, the radiation **106** may be directed from the intermediate focus **116** and toward the mirror **130a** (e.g., by the radiation source **102**). The radiation **106** incident upon the mirror **130a** (or a portion thereof) is reflected off the mirror facets **202**. The reflected radiation **208** is directed toward the pixels **204** of the illumination system **130b** as a plurality of beams. Each respective beam of the reflected radiation **208** is incident upon one or more pixels **204** of the illumination system **130b**. Radiation is reflected off the pixels **204** and toward the reticle **108** (or other intervening mirrors). In some implementations, the reflected radiation **208** includes unpolarized EUV radiation, and the pixels **204** are configured to modify the unpolarized EUV radiation in various ways including polarizing the unpolarized EUV radiation. Thus, the EUV radiation reflected by the pixels **204** may include TE polarized EUV radiation, TM polarized EUV radiation, unpolarized EUV radiation, or a combination thereof.

(27) As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

(28) FIG. 3 is a diagram of an example implementation **300**. The example implementation **300** illustrates a plurality of polarization configurations for the pixels **204** of the illumination system **130b** described herein for use in the lithography system of FIG. 1. As shown in FIG. 3, the pixels **204** are arranged in a grid pattern on the substrate **206**. However, the pixels **204** may be arranged in another pattern, such as a staggered pattern (e.g., a brick pattern), an asymmetric pattern, a non-uniform pattern, and/or another type of pattern.

(29) As further shown in FIG. 3, the pixels **204** are configured to reflect a particular type of radiation (e.g., EUV radiation). In particular, a pixel **204** may be configured as an unpolarized pixel **302** (e.g., in an unpolarized polarization configuration), a TE polarized pixel **304** (e.g., in a TE polarized polarization configuration), or a TM polarized pixel **306** (e.g., in a TM polarized polarization configuration). In some implementations, a pixel **204** may be configured as a plurality of polarization types (e.g., TE polarization and TM polarization), as described herein.

(30) An unpolarized pixel **302** includes a pixel **204** that is configured to reflect unpolarized radiation (e.g., unpolarized EUV radiation). Unpolarized pixels **302** may be capable of reflecting a greater intensity of EUV radiation relative to polarized pixels, which enables the unpolarized pixels **302** to increase the throughput of the exposure tool **104**. The arrows of the unpolarized pixels **302** represent the non-specific and non-directional attributes of the unpolarized radiation.

(31) A TE polarized pixel **304** includes a pixel **204** that is configured to reflect TE polarized radiation (e.g., TE polarized EUV radiation). TE polarized radiation refers to electromagnetic radiation (or light) in which the electric field of the electromagnetic radiation is normal (or perpendicular) to the plane of incidence of the electromagnetic radiation, and in which the magnetic field of the electromagnetic radiation is along (or parallel to) the plane of incidence. TE polarized pixels **304** may reflect EUV radiation at a lower intensity relative to an unpolarized pixel **302** because the TE polarized radiation is only one component of unpolarized radiation—the other component being TM polarized radiation. However, TE polarized radiation may increase the contrast of a pattern transferred from the reticle **108** to the semiconductor substrate **110** by the reflected radiation (particularly at higher numerical apertures). The increased contrast is provided by the complete (or near complete) destructive interference of the TE polarized radiation, which results in a final electric vector of 0 at the semiconductor substrate **110**. In other words, the TE polarized radiation is brighter (e.g., greater intensity) in the constructive interference of the TE polarized radiation and darker (or completely dark) in the destructive interference of the TE polarized radiation.

(32) A TM polarized pixel **306** includes a pixel **204** that is configured to reflect TM polarized radiation (e.g., TM polarized EUV radiation). TM polarized radiation refers to electromagnetic radiation (or light) in which the electric field of the electromagnetic radiation is along (or parallel to) the plane of incidence of the electromagnetic radiation, and in which the magnetic field of the electromagnetic radiation is normal (or perpendicular) to the plane of incidence.

(33) The pixels **204** may be configured in various combinations and/or arrangements of polarization configurations to achieve particular types of polarization patterns for the illumination system **130b**. For example, the pixels **204** of the illumination **130b** may include a combination of unpolarized pixels **302**, TE polarized pixels **304**, and TM polarized pixels **306** to achieve a radial polarization pattern or an azimuthal polarization pattern, among other examples. Moreover, in some implementations, one or more of the pixels **204** are configurable in that one or more of the pixels **204** are capable of switching between various polarization configurations, which enables free-form or arbitrary polarization of EUV radiation in the exposure tool **104**. In other words, this provides the exposure tool **104** (and the controller **140**) with the flexibility to optimize polarization patterns for different exposure operations, different pattern configurations of reticles **108** used in the exposure tool **104**. The quantity of unpolarized pixels **302** included on the substrate **206** may be increased to increase the reflectivity of the illumination system **130b** and to increase the throughput of the exposure tool **104**, or may be decreased to enable a greater quantity of TE polarized pixels **304** and/or a greater quantity of TM polarized pixels **306** to be included on the substrate **206**. The quantity of TE polarized pixels **304** may be increased to increase the contrast of the pattern transferred from the reticle **108** to the semiconductor substrate **110**, or may be decreased to enable a great quantity of unpolarized pixels **302** and/or a greater quantity of TM polarized pixels **306** to be included on the substrate **206**. The quantity of TM polarized pixels **306** may be increased to enable flexibility in configuring particular types of polarization patterns, or may be decreased to enable a great quantity of unpolarized pixels **302** and/or a greater quantity of TE polarized pixels **304** to be included on the substrate **206**.

(34) As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

(35) FIG. 4 is a side view diagram of an example implementation **400** of an unpolarized pixel **302** described herein. In particular, the example implementation **400** illustrates the structure of the unpolarized pixel **302** and the operation of the unpolarized pixel **302**.

(36) As shown in FIG. 4, the unpolarized pixel **302** includes a multilayer mirror (ML mirror) **402**. The multilayer mirror **402** may physically occupy the entire approximate area of the unpolarized pixel **302** or a portion thereof. The multilayer mirror **402** includes a base layer **404** and a plurality of alternating layers over and/or on the base layer **404**. The alternating layers include a plurality of layers **406** and a plurality of layers **408**, where a layer **406** is included over and/or on the base layer **404**, a layer **408** is included over and/or on the layer **406**, another layer **406** is included over and/or on the layer **408**, another layer **408** is included over and/or on the other layer **406**, and so on. In some implementations, the layers **406** and **408** are formed as a coating on the base layer **404**. In some implementations, the layers **406** and **408** are formed as a separate structure that is subsequently bonded to the base layer **404**.

(37) The layers **406** and **408** include alternating layers of molybdenum and silicon (Mo/Si layers), molybdenum and beryllium (Mo/Be layers), or another combination of layers that have different refractive indices. The combination of the materials in the layers **406** and **408** may be selected to provide a difference in refractive indices between the layers **406** and **408** (e.g., to provide reflectivity at an interface of the layers **406** and **408** according to Fresnel's equations), while reducing and/or minimizing extinction coefficients for the layers **406** and **408** (e.g., to minimize absorption).

(38) In general, the reflectivity of the multilayer mirror **402** may increase as a quantity of pairs of the layers **406** and **408** is increased. In some implementations, the multilayer mirror **402** includes 20 to 40 pairs of the layers **406** and **408**, which enables the multilayer mirror **402** to achieve a reflectivity of approximately 60% to approximately 80%. However, other quantities of pairs of the layers **406** and **408** are within the scope of the present disclosure. A thickness of the silicon layers (e.g., the layers **408**) may be included in a range of approximately 2 nanometers (nm) to about 6 nm, and a thickness of the molybdenum layers (e.g., the layers **406**) may be included in a range of

approximately 1 nm to approximately 5 nm to achieve suitable reflectivity and absorption performance. However, other values for the thickness of the layers **406** and **408** are within the scope of the present disclosure.

(39) As further shown in FIG. **4**, unpolarized radiation **208** (e.g., unpolarized EUV radiation reflected by the mirror facets **202** from the mirror **130a** toward the illumination system **130b**) incident upon the multilayer mirror **402** of the unpolarized pixel **302** is reflected as reflected unpolarized radiation (e.g., reflected unpolarized EUV radiation) **410** by the multilayer mirror **402**. It is noted that the FIG. **4** illustrates the direction of travel of the unpolarized radiation **208** (e.g., a light path of the unpolarized radiation **208**) and a direction of travel of the reflected unpolarized radiation **410** (e.g., a light path of the reflected unpolarized radiation **410**). In practice, the unpolarized radiation **208** may illuminate approximately the entire surface area of the multilayer mirror **402**, and the reflected unpolarized radiation **410** may be reflected off approximately the entire surface area of the multilayer mirror **402**. Moreover, portions of the reflected unpolarized radiation **410** may be reflected off one or more of the layers **406** and/or **408** further down from the top surface of the multilayer mirror **402**.

(40) The overall angle **412** between a ray of the unpolarized radiation **208** incident upon the multilayer mirror **402** and a ray of a corresponding reflected unpolarized radiation **410** may be referred to as a chief ray angle (CRA) or as a chief ray angle at object (CRAO). The overall angle **412** may be approximately 8 degrees to approximately 16 degrees or another angle. The overall angle **412** may include the sum of the angle of incidence **414a** of the ray of the unpolarized radiation **208** and the angle of reflectance **414b** of the ray of the reflected unpolarized radiation **410**. The magnitude of the angle of incidence **414a** and the magnitude of the angle of reflectance **414b** are relative to an axis **416** that is approximately perpendicular to the surface of reflection of the multilayer mirror **402**. In some implementations, the magnitude of the angle of incidence **414a** and the magnitude of the angle of reflectance **414b** are each approximately 4 to approximately 8 degrees. However, other values for the magnitude of the angle of incidence **414a** and the magnitude of the angle of reflectance **414b** are within the scope of the present disclosure.

(41) Note that TE and TM modes are defined relative to the plane of incidence in all examples in the present disclosure. In FIG. **4**, for example, the incident beam **208** comes from left side with incident angle **414a** to the axis **416**. Therefore, the electric field is in the Y direction for the TE polarized pixel **304**. The electric field is in the X direction for the TM polarized pixel **306**. The TE and TM mode are different when considering the imaging on a wafer with certain orientation of a particular pattern. For periodic line and space array in the Y direction, the TE mode is the one with electric field in the Y direction, as in the TE polarized pixel **304** in FIG. **3**, and the TM mode is with electric field in the X direction, as in the TM polarized pixel **306** in FIG. **3**. For periodic line and space array in X direction, the TE mode is with the electric field in the X direction, and the TM mode is with the electric field in the Y direction.

(42) Fig. As indicated above, FIG. **4** is provided as an example. Other examples may differ from what is described with regard to FIG. **4**.

(43) **5** is a side view diagram of an example implementation **500** of a TE polarized pixel **304** described herein. In particular, the example implementation **500** illustrates the structure of the TE polarized pixel **304** and the operation of the TE polarized pixel **304**.

(44) As shown in FIG. **5**, the TE polarized pixel **304** includes a multilayer mirror **402** as described above in connection with FIG. **4**. Moreover, the TE polarized pixel **304** includes a multilayer polarizer (e.g., an ML polarizer structure) **502**. The multilayer polarizer **502** includes a plurality of alternating layers, including a plurality of layers **504** and a plurality of layers **506**, where a layer **506** is included over and/or on a layer **504**, another layer **504** is included over and/or on the layer **506**, another layer **506** is included over and/or on the other layer **504**, and so on.

(45) The layers **504** and **506** include alternating layers of molybdenum and silicon (Mo/Si layers), molybdenum and beryllium (Mo/Be layers), or another combination of layers that have different

refractive indices. The combination of the materials in the layers **504** and **506** may be selected to provide a difference in refractive indices between the layers **504** and **506** (e.g., to provide reflectivity at an interface of the layers **504** and **506** according to Fresnel's equations), while providing reducing and/or minimizing extinction coefficients for the layers **504** and **506** (e.g., to minimize absorption). The quantity of pairs including a layer **504** and a layer **506** may be included in a range of 18 pairs to 22 pairs to provide sufficient reflectivity and sufficient polarization. However, other values for the quantity of the pairs are within the scope of the present disclosure.

(46) The thickness of a layer **504** and the thickness of a layer **506** may be different to achieve polarization of the unpolarized radiation **208** incident upon the multilayer polarizer **502**. In particular, the difference in the respective thicknesses of the layers **504** and the layers **506** facilitate the separation of the unpolarized radiation **208** into reflected TE polarized radiation **508** and transmitted TM polarized radiation. In particular, the reflected TE polarized radiation **508** is reflected off of the multilayer polarizer **502** and toward the multilayer mirror **402**, whereas the transmitted TM polarized radiation is transmitted through the multilayer polarizer **502** (e.g., and is not reflected by the TE polarized pixel **304**). In some implementations, the thickness of the layers **504** (e.g., which may include molybdenum layers) are included in a range of approximately 2.2 nm to approximately 2.8 nm, whereas the thickness of the layers **506** (e.g., which may include silicon layers) are included in a range of approximately 6.7 nm to approximately 7.3 nm to achieve a sufficient reflectance degree of polarization (DOP) and to achieve a sufficient transmittance degree of polarization. However, other values for the thicknesses of the layers **504** and **506** are within the scope of the present disclosure.

(47) As further shown in FIG. 5, the unpolarized radiation **208** incident upon the multilayer polarizer **502** of the TE polarized pixel **304** is reflected as reflected TE polarized radiation (e.g., reflected TE polarized EUV radiation) **508** by the multilayer polarizer **502** toward the multilayer mirror **402**. It is noted that the FIG. 5 illustrates the direction of travel of the unpolarized radiation **208** (e.g., a light path of the unpolarized radiation **208**) and a direction of travel of the reflected TE polarized radiation **508** (e.g., a light path of the reflected TE polarized radiation **508**). In practice, the unpolarized radiation **208** may illuminate approximately the entire surface area of the multilayer polarizer **502**, and the reflected TE polarized radiation **508** may be reflected onto approximately the entire surface area of the multilayer mirror **402** (or a portion thereof). Moreover, portions of the reflected TE polarized radiation **508** may be reflected off of one or more of the layers **504** and/or **506** further down from the top surface of the multilayer polarizer **502**, and may be reflected off of one or more of the layers **406** and/or **408** further down from the top surface of the multilayer mirror **402**.

(48) The overall angle **510** between a ray of the unpolarized radiation **208** incident upon the multilayer polarizer **502** and a ray of a corresponding reflected TE polarized radiation **508** may be approximately 8 degrees to approximately 16 degrees or another angle. The magnitude of the angle of incidence **512a** of the unpolarized radiation **208** toward the multilayer polarizer **502** relative to an axis **514** that is approximately perpendicular to the surface of reflection of the multilayer polarizer **502**, and the magnitude of the angle of reflectance **512b** of the reflected TE polarized radiation **508** reflected off of the multilayer polarizer **502**, may each be included in a range of approximately 40 degrees to approximately 44 degrees to achieve a high reflectance degree of polarization (e.g., approximately 99% reflectance degree of polarization or greater). However, other values for the magnitude of the angle of incidence **512a** and the magnitude of the angle of reflectance **512b** are within the scope of the present disclosure.

(49) The magnitude of the angle of incidence **516a** of the reflected TE polarized radiation **508** incident upon the multilayer mirror **402** relative to an axis **518** that is approximately perpendicular to the surface of reflection of the multilayer mirror **402**, and the magnitude of the angle of reflectance **516b** of the reflected TE polarized radiation **508** reflected off of the multilayer mirror **402**, may each be included in a range of approximately 52 degrees to approximately 56 degrees to

achieve a particular chief ray angle (e.g., the overall angle **510**) for the TE polarized pixel **304**. However, other values for the magnitude of the angle of incidence **512a** and the magnitude of the angle of reflectance **512b** are within the scope of the present disclosure. The chief ray angle may include the difference between the angle of reflectance **516b** and the angle of incidence **512a**. As an example, the chief ray angle may be approximately 12 degrees for a 42 degree angle of incidence **512a** and a 54 degree angle of reflectance **516b**. However, other values for the chief ray angle of the TE polarized pixel **304** are within the scope of the present disclosure.

(50) In some implementations, a combination of the parameters described above for the multilayer polarizer **502**, such as the quantity of alternating pairs of the layers **504** and **506**, the differences in thicknesses between the layers **504** and **506**, and the angle of incidence **512a** of the unpolarized radiation **208** toward the multilayer polarizer **502** may be configured to achieve or provide particular performance parameters for multilayer polarizer **502**. For example, a combination of the parameters for the multilayer polarizer **502** may be configured in one or more of the ranges described above (and/or other ranges) to achieve a reflectivity of TM polarized radiation of approximately 0.01% or less and a reflectivity of the TE polarized radiation of approximately 34% or greater. As another example, a combination of the parameters for the multilayer polarizer **502** may be configured in one or more of the ranges described above (and/or other ranges) to achieve a reflectance degree of polarization of approximately 99% or greater.

(51) As further shown in FIG. 5, the multilayer polarizer **502** may be positioned lower than the multilayer mirror **402** in the TE polarized pixel **304**. For example, the highest edge (or corner) of the multilayer polarizer **502** may be lower than the highest edge (or corner) of the multilayer mirror **402**, and/or the lowest edge (or corner) of the multilayer polarizer **502** may be lower than the lowest edge (or corner) of the multilayer mirror **402**. The lower relative position of the multilayer polarizer **502** enables the angle of incidence **512a** to be configured to achieve a high reflectance degree of polarization while enabling the angle of reflectance **516b** to be configured to achieve a particular chief ray angle for the TE polarized pixel **304**.

(52) As indicated above, FIG. 5 is provided as an example. Other examples may differ from what is described with regard to FIG. 5.

(53) FIG. 6 is a diagram of an example implementation **600** of a TM polarized pixel **306** described herein. In particular, the example implementation **600** illustrates the structure of the TM polarized pixel **306** and the operation of the TM polarized pixel **306**.

(54) As shown in FIG. 6, the TM polarized pixel **306** includes a multilayer mirror **402** as described above in connection with FIG. 4. Moreover, the TM polarized pixel **306** includes a multilayer polarizer **502** as described above in connection with FIG. 5.

(55) As further shown in FIG. 6, the multilayer polarizer **502** separates and/or extracts TM polarized radiation from unpolarized radiation **208** incident upon the multilayer polarizer **502**. In other words, the multilayer structure of the multilayer polarizer **502** separates the unpolarized radiation **208** into TE polarized radiation (not shown) and the TM polarized radiation, which passes through the multilayer polarizer **502** and is transmitted toward the multilayer mirror **402** as transmitted TM polarized radiation **602a**. The multilayer mirror **402** is positioned below and/or under the multilayer polarizer **502** to receive the transmitted TM polarized radiation **602a**. The multilayer mirror **402** reflects the transmitted TM polarized radiation **602a** as reflected TM polarized radiation **602b**.

(56) It is noted that the FIG. 6 illustrates the direction of travel of the unpolarized radiation **208** (e.g., a light path of the unpolarized radiation **208**), a direction of travel of the transmitted TM polarized radiation **602a** (e.g., a light path of the transmitted TM polarized radiation **602a**), and a direction of travel of the reflected TM polarized radiation **602b** (e.g., a light path of the reflected TM polarized radiation **602b**). In practice, the unpolarized radiation **208** may illuminate approximately the entire surface area of the multilayer polarizer **502**, and the transmitted TM polarized radiation **602a** may be transmitted onto approximately the entire surface area of the

multilayer mirror **402** (or a portion thereof). Moreover, portions of the reflected TM polarized radiation **602b** may be reflected off one or more of the layers **406** and/or **408** further down from the top surface of the multilayer mirror **402**.

(57) The overall angle **604** between a ray of the unpolarized radiation **208** incident upon the multilayer polarizer **502** and a ray of a corresponding reflected TM polarized radiation **602b** may be approximately 8 degrees to approximately 16 degrees or another angle. The magnitude of the angle of incidence **606** of the unpolarized radiation **208** toward the multilayer polarizer **502** relative to an axis **608** that is approximately perpendicular to the surface of reflection of the multilayer polarizer **502** may be included in a range of approximately 40 degrees to approximately 44 degrees to achieve a high transmission degree of polarization (e.g., approximately 85% transmittance degree of polarization or greater) and to achieve a high degree of transmittance (e.g., approximately 25% or greater). However, other values for the magnitude of the angle of incidence **606** are within the scope of the present disclosure.

(58) The magnitude of the angle of incidence **610a** of the transmitted TM polarized radiation **602a** and the magnitude of the angle of reflectance **610b** of the reflected TM polarized radiation **602b** are relative to an axis **612** that is approximately perpendicular to the surface of reflection of the multilayer mirror **402**. In some implementations, the magnitude of the angle of incidence **610a** and the magnitude of the angle of reflectance **610b** are each approximately 4 to approximately 8 degrees. However, other values for the magnitude of the angle of incidence **610a** and the magnitude of the angle of reflectance **610b** are within the scope of the present disclosure.

(59) In some implementations, a combination of the parameters described above for the multilayer polarizer **502**, such as the quantity of alternating pairs of the layers **504** and **506**, the differences in thicknesses between the layers **504** and **506**, and the angle of incidence **606** of the unpolarized radiation **208** toward the multilayer polarizer **502** may be configured to achieve or provide particular performance parameters for multilayer polarizer **502**. For example, a combination of the parameters for the multilayer polarizer **502** may be configured in one or more of the ranges described above (and/or other ranges) to achieve a reflectivity of TM polarized radiation of approximately 23% or greater and a transmittance of the TM polarized radiation of approximately 2% or less. As another example, a combination of the parameters for the multilayer polarizer **502** may be configured in one or more of the ranges described above (and/or other ranges) to achieve a transmittance degree of polarization of approximately 85% or greater.

(60) As indicated above, FIG. **6** is provided as an example. Other examples may differ from what is described with regard to FIG. **6**.

(61) FIG. **7** is a diagram of an example implementation **700** of a configurable pixel **702** described herein. The example pixel configurations described above in connection with FIGS. **4-6** may be referred to as fixed pixel configurations in that a pixel **204** is configured in one of the pixel configurations described above in connection with FIGS. **4-6** and is not changed. As an example, a pixel **204** may be configured as an unpolarized pixel **302** as illustrated and described above in connection with FIG. **4**, where the configuration of the pixel **204** does not change from the configuration as an unpolarized pixel **302**. As another example, a pixel **204** may be configured as a TE polarized pixel **304** as illustrated and described above in connection with FIG. **5**, where the configuration of the pixel **204** does not change from the configuration as a TE polarized pixel **304**. As another example, a pixel **204** may be configured as a TM polarized pixel **306** as illustrated and described above in connection with FIG. **6**, where the configuration of the pixel **204** does not change from the configuration as a TM polarized pixel **306**. The configurable pixel **702** illustrated in FIG. **7** is configurable in that the polarization configuration of the configurable pixel **702** is capable of being selectively changed to different polarization configurations, as described herein. The capability to change polarization configuration for configurable pixels **702** on the illumination system **130b** enables free-form polarization for the exposure tool **104**. Moreover, the configurable pixel **702** is capable of reflecting separated types of polarized radiation (e.g., at the same time),

including TE polarized radiation and TM polarized radiation.

(62) The configurable pixel **702** (or a plurality of configurable pixels **702**) may be included on the substrate **206** of the illumination system **130b** (e.g., as pixel(s) **204**). In some implementations, the illumination system **130b** includes all configurable pixels **702**, which increases the flexibility of the illumination system **130b** to provide free-form polarization. In some implementations, the illumination system **130b** includes a combination of configurable pixels **702** and fixed pixels (e.g., pixels **204** that are each fixed in one of the polarization configurations illustrated in FIGS. **4-6**) to provide a degree of free-form polarization while reducing the manufacturing complexity of the illumination system **130b**.

(63) As shown in FIG. **7**, the configurable pixel **702** includes a multilayer polarizer **502** and a plurality of multilayer mirrors **402**, including a multilayer mirror **402a** under and/or below the multilayer polarizer **502** and another multilayer mirror **402b** adjacent to the multilayer polarizer **502**. The multilayer polarizer **502** may receive unpolarized radiation **208**, may separate the unpolarized radiation **208** (e.g., using the plurality of pairs of alternating layers **504** and **506**) into transmitted TM polarized radiation **704a** and reflected TE polarized radiation **706**. The transmitted TM polarized radiation **704a** is reflected off the multilayer mirror **402a**, which provides reflected TM polarized radiation **704b**. The reflected TE polarized radiation **706** is reflected off the multilayer mirror **402b**, which reflects the TE polarized radiation **706**. In the example implementation **700** illustrated in FIG. **7**, the configurable pixel **702** is configured in a polarization configuration in which the configurable pixel **702** reflects the reflected TM polarized radiation **704a** in a first portion of the configurable pixel **702** and reflects the reflected TE polarized radiation **706** in a second (adjacent) portion of the configurable pixel **702**.

(64) The overall angle **708** may correspond to the overall angle **604**. The overall angle **710** may correspond to the overall angle **510**. The angle of incidence **712a** relative to the axis **714** may correspond to the angle of incidence **512a** and/or the angle of incidence **606**. The angle of reflectance **712b** relative to the axis **714** may correspond to the angle of reflectance **512b**. The angle of incidence **716a** relative to the axis **718** may correspond to the angle of incidence **414a** and/or the angle of incidence **610a**. The angle of reflectance **716b** relative to the axis **718** may correspond to the angle of reflectance **414b** and/or the angle of reflectance **610b**. The angle of incidence **720a** relative to the axis **722** may correspond to the angle of incidence **516a**. The angle of reflectance **720b** relative to the axis **722** may correspond to the angle of reflectance **516b**.

(65) As further shown in FIG. **7**, the multilayer polarizer **502**, the multilayer mirror **402a**, and the multilayer mirror **402b** may each include a respective actuator such that the multilayer polarizer **502**, the multilayer mirror **402a**, and the multilayer mirror **402b** may be independently actuated to change the polarization configuration of the configurable pixel **702**. For example, the multilayer polarizer **502** includes an actuator **724a**, the multilayer mirror **402a** includes an actuator **724b**, and the multilayer mirror **402b** includes an actuator **724c**. The actuators **724a-724c** may include a servo-controlled motor, a direct current (DC) brushed or brushless motor, a linear motor, a stepper motor, and/or another type of actuator that is capable of rotating, translating, and/or displacing the multilayer polarizer **502**, the multilayer mirror **402a**, and/or the multilayer mirror **402b**.

(66) The controller **140** may communicate with the actuators **724a-724c** to receive sensor data associated with the multilayer polarizer **502**, the multilayer mirror **402a**, and/or the multilayer mirror **402b**. The sensor data may include or may indicate, for example, position information associated with the multilayer polarizer **502**, the multilayer mirror **402a**, and/or the multilayer mirror **402b**, rotational velocity and/or linear velocity of the multilayer polarizer **502**, the multilayer mirror **402a**, and/or the multilayer mirror **402b**, and/or other types of sensor data. The controller **140** is further configured to communicate with the actuators **724a-724c** to provide one or more signals to the actuators **724a-724c** to selectively configure the configurable pixel **702** in various polarization configurations described herein. The signal(s) may include analog signals (e.g., a voltage, a current), digital signals (e.g., digital communications), and/or other types of

signals that are transmitted over a wired and/or wireless connection to the actuators **724a-724c**. The controller **140** may provide or transmit a signal to the actuator **724a** to cause the actuator **724a** to actuate the multilayer polarizer **502** to cause the configurable pixel **702** to be configured in a particular polarization configuration. The controller **140** may provide or transmit a signal to the actuator **724b** to cause the actuator **724b** to actuate the multilayer mirror **402a** to cause the configurable pixel **702** to be configured in a particular polarization configuration. The controller **140** may provide or transmit a signal to the actuator **724c** to cause the actuator **724c** to actuate the multilayer mirror **402b** to cause the configurable pixel **702** to be configured in a particular polarization configuration.

(67) In some implementations, the controller **140** provides or transmits signals to one or more actuators of a plurality of configurable pixels **702** included on the illumination system **130b** to cause the illumination system **130b** to be configured in a particular polarization pattern or to enable free-form polarization. In some implementations, the controller **140** provides or transmits signals to one or more actuators of a plurality of configurable pixels **702** to cause subsets of the plurality of configurable pixels **702** to be configured in respective pixel configurations. For example, the controller **140** may provide or transmit a first signal (or a first set of signals) to cause a first subset of configurable pixels **702** to be configured in a TE polarized configuration (e.g., as illustrated and described in connection with FIG. **8B**), may provide or transmit a second signal (or a second set of signals) to cause a second subset of configurable pixels **702** to be configured in a TM polarized configuration (e.g., as illustrated and described in connection with FIG. **8C**), and may provide or transmit a third signal (or a third set of signals) to cause a third subset of configurable pixels **702** to be configured in an unpolarized configuration (e.g., as illustrated and described in connection with FIG. **8A**). Additionally and/or alternatively, the controller **140** may provide or transmit a signal (or a set of signals) to cause a subset of configurable pixels **702** to be configured in the TE polarized and TM polarized configuration illustrated in FIG. **7**. In some implementations, the controller **140** provides or transmits a signal (or a set of signals) to cause a subset of configurable pixels **702** to be configured in an off configuration (e.g., as illustrated and described in connection with FIG. **8D**) such that the subset of configurable pixels **702** is deactivated (e.g., does not reflect incident light).

(68) In some implementations, the controller **140** determines the polarization pattern for the illumination system **130b** (or determines the specific combination of polarization configurations for the plurality of configurable pixels **702**) based on one or more parameters. The one or more parameters may be associated with an exposure operation of the semiconductor substrate **110** or may be associated with a substrate lot including a plurality of semiconductor substrates **110** that are to be processed in similar exposure operations. The one or more parameters may include, for example, the pattern of the reticle **108** that is to be transferred to the semiconductor substrate **110** (or the substrate lot), a contrast parameter for the exposure operation, a throughput parameter for the exposure operation, and/or other parameters.

(69) In some implementations, the controller **140** determines the polarization pattern for the illumination system **130b** (or determines the specific combination of polarization configurations for the plurality of configurable pixels **702**) using a machine learning model. The machine learning model may be trained on training data, which may include the specific combinations of polarization configurations for the plurality of configurable pixels **702** from a plurality of historical exposure operations (e.g., thousands of historical exposure operations or more), the throughput and/or contrast achieved for the historical exposure operations, the types of reticles **108** used for the historical exposure operations, and/or other types of parameters for the historical exposure operations. In some implementations, the controller **140** provides the desired outcomes for an exposure operation (e.g., the contrast parameter, the throughput parameters) along with the pattern of the reticle **108** that is to be used in the exposure operation, and the controller **140** uses the machine learning model to analyze combinations of polarization configurations for the plurality of configurable pixels **702** based on the training data to identify a specific combination of polarization

configurations for the plurality of configurable pixels **702** that are estimated to achieve the desired outcomes for the exposure operation. In some implementations, the controller **140** provides a candidate combination of polarization configurations for the plurality of configurable pixels **702** for an exposure operation along with the pattern of the reticle **108** that is to be used in the exposure operation, and the controller **140** uses the machine learning model to generate a likelihood or confidence that the candidate combination of polarization configurations for the plurality of configurable pixels **702** will achieve the desired outcomes for an exposure operation (e.g., the contrast parameter, the throughput parameters).

(70) In some implementations, the controller **140** determines the polarization configurations for the plurality of configurable pixels **702** prior to an exposure operation and transmits signals to cause the plurality of configurable pixels **702** to be configured in the polarization configurations prior to the exposure operation. Alternatively, or after the start of the exposure operation, the controller **140** may determine modified polarization configurations for one or more of the plurality of configurable pixels **702** during the exposure operation and transmit signals to cause the one or more configurable pixels **702** to be configured in the polarization configurations during the exposure operation. This “on-the-fly” polarization adjustment may enable the controller **140** to fine-tune and/or improve patterning performance during the exposure operation.

(71) As indicated above, FIG. **7** is provided as an example. Other examples may differ from what is described with regard to FIG. **7**.

(72) FIGS. **8A-8D** are diagrams of example implementations of configurable pixels **702** described herein. FIG. **8A** illustrates an example implementation **810** in which a configurable pixel **702** is configured in an unpolarized configuration. As shown in FIG. **8A**, the controller **140** transmits one or more signals to the actuator **724a** to cause the actuator **724a** to actuate the multilayer polarizer **502**. The actuation may include rotating the multilayer polarizer **502** as shown in FIG. **8A**, translating the multilayer polarizer **502**, and/or displacing the multilayer polarizer **502**. The actuator **724a** may actuate the multilayer polarizer **502** to move the multilayer polarizer **502** out of the light path of the unpolarized radiation **208** incident upon the multilayer polarizer **502**. As a result, the unpolarized radiation **208** instead is incident upon the multilayer mirror **402a**, which reflects the unpolarized radiation **208**. In this way, the multilayer polarizer **502** is moved out of the light path of the unpolarized radiation **208** such that the unpolarized radiation **208** remains unpolarized (e.g., the unpolarized radiation **208** is not split into TE polarized radiation and TM polarized radiation components). In some implementations, the controller **140** transmits another signal (or signals) to the multilayer mirror **402a** to cause the actuator **724b** to actuate the multilayer mirror **402a** to position the multilayer mirror **402a** to reflect the unpolarized radiation **208**.

(73) FIG. **8B** illustrates an example implementation **820** in which a configurable pixel **702** is configured in a TE polarized configuration. As shown in FIG. **8B**, the controller **140** transmits one or more signals to the actuator **724b** to cause the actuator **724b** to actuate the multilayer mirror **402a**. The actuation may include rotating the multilayer mirror **402a** as shown in FIG. **8B**, translating the multilayer mirror **402a**, and/or displacing the multilayer mirror **402a**. The actuator **724b** may actuate the multilayer mirror **402a** to angle the multilayer mirror **402a** such that the multilayer mirror **402a** directs the transmitted TM polarized radiation **704a** in a direction so as to refrain from providing the reflected TM polarized radiation **704b**. In this way, only the reflected TE polarized radiation **706** is provided by the configurable pixel **702** in the TE polarized configuration of the example implementation **820**. In some implementations, the controller **140** transmits another signal (or signals) to the multilayer polarizer **502** and/or to the multilayer mirror **402b** to cause the multilayer polarizer **502** and/or to the multilayer mirror **402b** to reflect the reflected TE polarized radiation **706**.

(74) FIG. **8C** illustrates an example implementation **830** in which a configurable pixel **702** is configured in a TM polarized configuration. As shown in FIG. **8C**, the controller **140** transmits one or more signals to the actuator **724c** to cause the actuator **724c** to actuate the multilayer mirror

402b. The actuation may include rotating the multilayer mirror **402b** as shown in FIG. 8C, translating the multilayer mirror **402b**, and/or displacing the multilayer mirror **402b**. The actuator **724c** may actuate the multilayer mirror **402b** to angle the multilayer mirror **402b** such that the multilayer mirror **402b** directs the reflected TE polarized radiation **706** in a direction so as to refrain from providing the reflected TE polarized radiation **706**. In this way, only the reflected TM polarized radiation **704b** is provided by the configurable pixel **702** in the TM polarized configuration of the example implementation **830**. In some implementations, the controller **140** transmits another signal (or signals) to the multilayer polarizer **502** and/or to the multilayer mirror **402a** to cause the multilayer polarizer **502** and/or to the multilayer mirror **402a** to reflect the reflected TM polarized radiation **704b**.

(75) FIG. 8D illustrates an example implementation **840** in which a configurable pixel **702** is configured in a deactivated configuration. In the deactivated configuration, the configurable pixel **702** refrains from providing radiation output. As shown in FIG. 8D, the controller **140** transmits one or more signals to the actuator **724a** to cause the actuator **724a** to actuate the multilayer polarizer **502**. This causes the actuator **724a** to move the multilayer polarizer **502** out of the light path of the unpolarized radiation **208** incident upon the multilayer polarizer **502**. Moreover, the controller **140** transmits one or more signals to the actuator **724b** to cause the actuator **724b** to actuate the multilayer mirror **402a**. This causes the actuator **724b** to move the multilayer mirror **402a** such that the multilayer mirror **402a** is configured at an angle in which the multilayer mirror **402a** directs the unpolarized radiation **208** such that no output of the configurable pixel **702** is provided.

(76) As indicated above, FIGS. 8A-8D are provided as examples. Other examples may differ from what is described with regard to FIGS. 8A-8D.

(77) FIG. 9 is a diagram of an example implementation **900** of a configurable pixel **702** described herein. The example implementation **900** of a configurable pixel **702** is similar to the example implementation **700** of a configurable pixel **702** described above in connection with FIG. 7. In addition, the configurable pixel **702** in the example implementation **900** includes a mirror **902** that is positioned in the light path of the reflected TE polarized radiation **706** between the multilayer polarizer **502** and the multilayer mirror **402b**. The mirror **902** includes a tunable mirror, a single layer mirror, a multilayer mirror, a mirror including a coating formed on a substrate, or another type of mirror. The mirror **902** enables multilayer mirror **402b** to be positioned further away from the multilayer polarizer **502**, which in turn enables the reflected TE polarized radiation **706** to be provided to different locations (e.g., further locations than without the use of the mirror **902**) on the reticle **108**. Moreover, while the multilayer mirror **402b** is illustrated in the example implementation **900** as being included in the configurable pixel **702**, in other implementations, the mirror **902** may be configured to provide the reflected TE polarized radiation **706** to a multilayer mirror **402** included in another pixel **204**. In this way, radiation from a plurality of pixels **204** can be combined to further increase the polarization flexibility of the illumination system **130b**.

(78) The magnitude of the angle of incidence **904a** of the TE polarized radiation **706** toward the multilayer mirror **402b** relative to an axis **906** that is approximately perpendicular to the surface of reflection of the mirror **902**, and the magnitude of the angle of reflectance **904b** of the reflected TE polarized radiation **706** reflected off of the mirror **902**, may each be included in a range of approximately 75 degrees to approximately 81 degrees to achieve a particular angle for the angle of incidence **720a** toward the multilayer mirror **402b** and/or to achieve a particular chief ray angle for the configurable pixel **702**. However, other values for the angle of incidence **904a** and for the angle of reflectance **904b** are within the scope of the present disclosure.

(79) As further shown in FIG. 9, an actuator **908** may be included on the mirror **902** such that the mirror **902** may be actuated in addition to (or alternatively to) the multilayer polarizer **502**, the multilayer mirror **402a**, and/or the multilayer mirror **402b**. The controller **140** may communicate with the actuator **908** to cause the actuator **908** to actuate the mirror **902** to change or modify the

polarization of the configurable pixel **702**. Moreover, the controller **140** may communicate with the actuator **908** to cause the actuator **908** to actuate the mirror **902** such that the mirror **902** provides the reflected TE polarized radiation **706** to a multilayer mirror **402** in a particular pixel **204** of the illumination system **130b**. In this way, the mirror **902** may be actuated to selectively direct the reflected TE polarized radiation **706** to different pixels **204**, which further increases the polarization flexibility of the illumination system **130b**.

(80) As indicated above, FIG. **9** is provided as an example. Other examples may differ from what is described with regard to FIG. **9**.

(81) FIGS. **10A-10C** are diagrams of example pixel configurations for the illumination system **130b** described herein for use in the lithography system **100** of FIG. **1**. The example pixel configurations described in connection with FIGS. **10A-10C** enable the illumination system **130b** to achieve and/or provide particular types of polarized EUV radiation (and/or other types of polarized radiation). The illumination system **130b** may be configured in one or more of the example pixel configurations described in connection with FIGS. **10A-10C** using fixed polarized pixels (e.g., one or more of the pixels **302**, **304**, and/or **306**), configurable pixels **702**, and/or a combination thereof. In some implementations, illumination system **130b** includes configurable pixels **702** and is capable of dynamically switching between pixel configurations (and thus, polarization configurations), which enables the illumination system **130b** to provide free-form polarization.

(82) FIG. **10A** illustrates an example pixel configuration **1010**. In the example pixel configuration **1010**, the pixels **204** are configured in a TE polarized configuration. The pixels **204** may be configured as fixed TE polarized pixels (e.g., TE polarized pixels **304** as illustrated in FIG. **5**), as configurable pixels **702** that are configured in a TE polarized configuration as illustrated in FIG. **8B**, or a combination thereof. In some implementations, the controller **140** transmits a signal to the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the pixels **204** in the TE polarized configuration.

(83) In the example pixel configuration **1010**, the illumination system **130b** is configured to generate and/or provide linear TE polarized radiation. Accordingly, the example pixel configuration **1010** may be referred to as a linear TE polarization configuration. In some implementations, the pixels **204** are configured to provide other types of linear polarized radiation such as linear TM polarized radiation.

(84) FIG. **10B** illustrates an example pixel configuration **1020**. In the example pixel configuration **1020**, a first subset of the pixels **204** are configured in a TE polarized configuration, a second subset of the pixels **204** are configured in a TM polarized configuration, and a third subset of the pixels **204** are configured in an unpolarized configuration. The first subset of the pixels **204** may be configured as fixed TE polarized pixels (e.g., TE polarized pixels **304** as illustrated in FIG. **5**), as configurable pixels **702** that are configured in a TE polarized configuration as illustrated in FIG. **8B**, or a combination thereof. The second subset of the pixels **204** may be configured as fixed TM polarized pixels (e.g., TM polarized pixels **306** as illustrated in FIG. **6**), as configurable pixels **702** that are configured in a TM polarized configuration as illustrated in FIG. **8C**, or a combination thereof. The third subset of the pixels **204** may be configured as fixed unpolarized pixels (e.g., unpolarized pixels **304** as illustrated in FIG. **4**), as configurable pixels **702** that are configured in an unpolarized configuration as illustrated in FIG. **8A**, or a combination thereof.

(85) In the example pixel configuration **1020**, the illumination system **130b** is configured to generate and/or provide radial polarized radiation. Accordingly, the example pixel configuration **1020** may be referred to as a radial polarization configuration. In some implementations, the controller **140** transmits a first signal to the first subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the first subset of the pixels **204** in the TE polarized configuration. In some implementations, the controller **140** transmits a second signal to the second subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or

908 (or a subset thereof) to configure the second subset of the pixels **204** in the TM polarized configuration. In some implementations, the controller **140** transmits a third signal to the third subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the third subset of the pixels **204** in the unpolarized configuration.

(86) FIG. **10C** illustrates an example pixel configuration **1030**. In the example pixel configuration **1030**, a first subset of the pixels **204** are configured in a TE polarized configuration, a second subset of the pixels **204** are configured in a TM polarized configuration, and a third subset of the pixels **204** are configured in an unpolarized configuration. The first subset of the pixels **204** may be configured as fixed TE polarized pixels (e.g., TE polarized pixels **304** as illustrated in FIG. **5**), as configurable pixels **702** that are configured in a TE polarized configuration as illustrated in FIG. **8B**, or a combination thereof. The second subset of the pixels **204** may be configured as fixed TM polarized pixels (e.g., TM polarized pixels **306** as illustrated in FIG. **6**), as configurable pixels **702** that are configured in a TM polarized configuration as illustrated in FIG. **8C**, or a combination thereof. The third subset of the pixels **204** may be configured as fixed unpolarized pixels (e.g., unpolarized pixels **304** as illustrated in FIG. **4**), as configurable pixels **702** that are configured in an unpolarized configuration as illustrated in FIG. **8A**, or a combination thereof.

(87) In the example pixel configuration **1030**, the illumination system **130b** is configured to generate and/or provide azimuthal polarized radiation. Accordingly, the example pixel configuration **1030** may be referred to as an azimuthal polarization configuration. In some implementations, the controller **140** transmits a first signal to the first subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the first subset of the pixels **204** in the TE polarized configuration. In some implementations, the controller **140** transmits a second signal to the second subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the second subset of the pixels **204** in the TM polarized configuration. In some implementations, the controller **140** transmits a third signal to the third subset of the pixels **204** to cause the associated actuators **724a-724c**, and/or **908** (or a subset thereof) to configure the third subset of the pixels **204** in the unpolarized configuration.

(88) It is noted that the quantity of pixels **204** included in the illumination system **130b** illustrated and described in the implementations herein are examples, and other quantities of pixels **204** may be included in the illumination system **130b**. Similarly, it is noted that the pixel configurations described herein for the illumination system **130b** are examples, and other pixel configurations are within the scope of the present disclosure.

(89) As indicated above, FIGS. **10A-10C** are provided as examples. Other examples may differ from what is described with regard to FIGS. **10A-10C**.

(90) FIG. **11** is a diagram of example components of a device **1100**, which may correspond to the controller **140**, the illumination system **130b**, one or more of the actuators **724a-724c**, and/or another component. In some implementations, the lithography system **100**, the radiation source **102**, the exposure tool **104**, the controller **140**, the illumination system **130b**, one or more of the actuators **724a-724c**, and/or another component may include one or more devices **1100** and/or one or more components of device **1100**. As shown in FIG. **11**, device **1100** may include a bus **1110**, a processor **1120**, a memory **1130**, an input component **1140**, an output component **1150**, and a communication component **1160**.

(91) Bus **1110** includes one or more components that enable wired and/or wireless communication among the components of device **1100**. Bus **1110** may couple together two or more components of FIG. **11**, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. Processor **1120** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **1120** is implemented in hardware, firmware, or a combination of hardware and software.

In some implementations, processor **1120** includes one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

(92) Memory **1130** includes volatile and/or nonvolatile memory. For example, memory **1130** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory). Memory **1130** may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). Memory **1130** may be a non-transitory computer-readable medium. Memory **1130** stores information, instructions, and/or software (e.g., one or more software applications) related to the operation of device **1100**. In some implementations, memory **1130** includes one or more memories that are coupled to one or more processors (e.g., processor **1120**), such as via bus **1110**.

(93) Input component **1140** enables device **1100** to receive input, such as user input and/or sensed input. For example, input component **1140** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, an accelerometer, a gyroscope, and/or an actuator. Output component **1150** enables device **1100** to provide output, such as via a display, a speaker, and/or a light-emitting diode. Communication component **1160** enables device **1100** to communicate with other devices via a wired connection and/or a wireless connection. For example, communication component **1160** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

(94) Device **1100** may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **1130**) may store a set of instructions (e.g., one or more instructions or code) for execution by processor **1120**. Processor **1120** may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors **1120**, causes the one or more processors **1120** and/or the device **1100** to perform one or more operations or processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, processor **1120** may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(95) The number and arrangement of components shown in FIG. **11** are provided as an example. Device **1100** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **11**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **1100** may perform one or more functions described as being performed by another set of components of device **1100**.

(96) FIG. **12** is a flowchart of an example process **1200** associated with configuring a pixel of an illumination system described herein. In some implementations, one or more process blocks of FIG. **12** may be performed by a controller (e.g., the controller **140**, the device **1100**). In some implementations, one or more process blocks of FIG. **12** may be performed by another device or a group of devices separate from or including the controller, such as a lithography system (e.g., the lithography system **100**), a radiation source (e.g., the radiation source **102**), an exposure tool (e.g., the exposure tool **104**), and/or an actuator (e.g., an actuator **724a-724c**, **908**), among other examples. Additionally, or alternatively, one or more process blocks of FIG. **12** may be performed by one or more components of device **1100**, such as processor **1120**, memory **1130**, input component **1140**, output component **1150**, and/or communication component **1160**.

(97) As shown in FIG. **12**, process **1200** may include transmitting a first signal to cause at least one of a plurality of configurable pixels of an illumination system to be configured in a polarization configuration of a plurality of polarization configurations (block **1210**). For example, the controller **140** may transmit a first signal to cause at least one of a plurality of configurable pixels **702** of the illumination system **130b** to be configured in a polarization configuration of a plurality of

polarization configurations (e.g., the polarization configurations described in connection with the example implementations **700**, **810-840**, and/or **900**), as described above.

(98) As further shown in FIG. **12**, process **1200** may include transmitting a second signal to cause a lithography system to perform an exposure operation while the at least one of the plurality of configurable pixels is configured in the polarization configuration (block **1220**). For example, the controller **140** may transmit a second signal to cause a lithography system **100** to perform an exposure operation while the at least one of the plurality of configurable pixels **702** is configured in the polarization configuration, as described above. In some implementations, the illumination system **130b** is included in the exposure tool **104** of the lithography system **100**.

(99) Process **1200** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(100) In a first implementation, transmitting the first signal includes transmitting the first signal to cause each of the plurality of configurable pixels **702** to be configured in respective polarization configurations prior to the exposure operation. In a second implementation, alone or in combination with the first implementation, transmitting the first signal includes transmitting the first signal to cause each of the plurality of configurable pixels **702** to be configured in respective polarization configurations during the exposure operation. In a third implementation, alone or in combination with one or more of the first and second implementations, process **1200** includes determining (e.g., by the controller **140**) a particular combination of respective polarization configurations for the plurality of configurable pixels **702** based on at least one of a contrast parameter for the exposure operation, or a throughput parameter for the exposure operation.

(101) In a fourth implementation, alone or in combination with one or more of the first through third implementations, transmitting the first signal includes transmitting the first signal to cause at least a subset of the plurality of configurable pixels **702** to be configured in a TE polarized configuration (e.g., as illustrated in the example implementation **820**). In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, transmitting the first signal includes transmitting the first signal to cause a first subset of the plurality of configurable pixels **702** to be configured in a TE polarized configuration (e.g., as illustrated in the example implementation **820**), a second subset of the plurality of configurable pixels **702** to be configured in a TM polarized configuration (e.g., as illustrated in the example implementation **830**), and a third subset of the plurality of configurable pixels **702** to be configured in an unpolarized configuration (e.g., as illustrated in the example implementation **810**). In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, transmitting the first signal includes transmitting the first signal to respective one or more actuators **724a-724c** and/or **908** associated with each of the plurality of configurable pixels **702** to cause the respective one or more actuators **724a-724c** and/or **908** to configure the plurality of configurable pixels **702** in the respective polarization configurations.

(102) Although FIG. **12** shows example blocks of process **1200**, in some implementations, process **1200** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **12**. Additionally, or alternatively, two or more of the blocks of process **1200** may be performed in parallel.

(103) FIG. **13** is a flowchart of an example process **1300** associated with configuring a pixel of an illumination system described herein. In some implementations, one or more process blocks of FIG. **13** may be performed by an illumination system (e.g., the illumination system **130b**, the device **1100**). In some implementations, one or more process blocks of FIG. **13** may be performed by another device or a group of devices separate from or including the illumination system, such as a lithography system (e.g., the lithography system **100**), a radiation source (e.g., the radiation source **102**), an exposure tool (e.g., the exposure tool **104**), a controller (e.g., the controller **140**), and/or an actuator (e.g., an actuator **724a-724c**), among other examples. Additionally, or

alternatively, one or more process blocks of FIG. 13 may be performed by one or more components of device 1100, such as processor 1120, memory 1130, input component 1140, output component 1150, and/or communication component 1160.

(104) As shown in FIG. 13, process 1300 may include configuring a plurality of multilayer mirrors and a multilayer polarizer of a configurable pixel included on a substrate of an illumination system (block 1310). For example, the illumination system 130b (e.g., the actuators 724a-724c and/or 908 of the illumination system 130b) may configure the multilayer mirrors 402a and 402b and the multilayer polarizer 502 of a configurable pixel 702 included on the substrate 206 of the illumination system 130b, as described above. In some implementations, the illumination system 130b is included in an EUV exposure tool (e.g., the exposure tool 104) of the lithography system 100. In some implementations, the plurality of multilayer mirrors 402a and 402b and the multilayer polarizer 502 are configured such that the configurable pixel 702 is configured in a particular polarization configuration (e.g., the polarization configurations described in connection with the example implementations 700, 810-840, and/or 900).

(105) As further shown in FIG. 13, process 1300 may include receiving unpolarized EUV radiation from a first mirror included in the EUV exposure tool during an exposure operation of the EUV exposure tool (block 1320). For example, the illumination system 130b may receive unpolarized EUV radiation (e.g., the unpolarized radiation 208) from a first mirror (e.g., the mirror 130a) included in the EUV exposure tool during an exposure operation of the EUV exposure tool, as described above.

(106) As further shown in FIG. 13, process 1300 may include providing reflected EUV radiation to a second mirror included in the EUV exposure tool during the exposure operation (block 1330). For example, the illumination system 130b may provide reflected EUV radiation (e.g., the unpolarized radiation 208, the reflected TM polarized radiation 704b, and/or the reflected TE polarized radiation 706) to a second mirror (e.g., the mirror 132) included in the EUV exposure tool during the exposure operation, as described above. In some implementations, a polarization of the reflected EUV radiation is based on the unpolarized EUV radiation and the particular polarization configuration of the configurable pixel 702.

(107) Process 1300 may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(108) In a first implementation, the particular polarization configuration includes a TE polarization configuration (e.g., as illustrated in the example implementation 820), and configuring the plurality of multilayer mirrors 402a and 402b and the multilayer polarizer 502 includes actuating a multilayer mirror (e.g., the multilayer mirror 402a), of the plurality of multilayer mirrors 402a and 402b, below the multilayer polarizer 502 such that the reflected EUV radiation includes a 1E polarized component (e.g., the reflected TE polarized radiation 706) of the unpolarized EUV radiation (e.g., the unpolarized radiation 208).

(109) In a second implementation, alone or in combination with the first implementation, providing the reflected EUV radiation includes polarizing, using the multilayer polarizer 502, the unpolarized EUV radiation (e.g., the unpolarized radiation 208) into the TE polarized component and a TM polarized component (e.g., the transmitted TM polarized radiation 704a), and reflecting, using another multilayer mirror (e.g., the multilayer mirror 402b) of the plurality of multilayer mirrors 402a and 402b, the TE polarized component toward the second mirror (e.g., the mirror 132), the other multilayer mirror (e.g., the multilayer mirror 402b) is adjacent to the multilayer polarizer 502, and the TM polarized component is directed away from the second mirror (e.g., the mirror 132) by the multilayer mirror (e.g., the multilayer mirror 402a).

(110) In a third implementation, alone or in combination with one or more of the first and second implementations, the particular polarization configuration includes a TM polarization configuration (e.g., as illustrated in the example implementation 830), and configuring the plurality of multilayer

mirrors **402a** and **402b** and the multilayer polarizer **502** includes actuating a multilayer mirror e.g., the multilayer mirror **402b**), of the plurality of multilayer mirrors **402a** and **402b**, adjacent to the multilayer polarizer **502** such that the reflected EUV radiation includes a TM polarized component (e.g., the reflected TM polarized radiation **704b**) of the unpolarized EUV radiation (e.g., the unpolarized radiation **208**).

(111) In a fourth implementation, alone or in combination with one or more of the first through third implementations, providing the reflected EUV radiation includes polarizing, using the multilayer polarizer **502**, the unpolarized EUV radiation (e.g., the unpolarized radiation **208**) into the TM polarized component and a TE polarized component (e.g., the reflected TE polarized radiation **706**), and reflecting, using another multilayer mirror (e.g., the multilayer mirror **402a**) of the plurality of multilayer mirrors **402a** and **402b**, the TM polarized component toward the second mirror (e.g., the mirror **132**), where the other multilayer mirror (e.g., the multilayer mirror **402a**) is below the multilayer polarizer **502**, and the TE polarized component is directed away from the second mirror (e.g., the mirror **132**) by the multilayer mirror e.g., the multilayer mirror **402b**).

(112) In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, the particular polarization configuration includes an unpolarized polarization configuration (e.g., as illustrated in the example implementation **810**), and configuring the plurality of multilayer mirrors **402a** and **402b** and the multilayer polarizer **502** includes actuating the multilayer polarizer **502** such that the reflected EUV radiation includes the unpolarized EUV radiation (e.g., the unpolarized radiation **208**).

(113) In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, providing the reflected EUV radiation includes reflecting, using a multilayer mirror (e.g., the multilayer mirror **402a**) of the plurality of multilayer mirrors **402a** and **402b**, the unpolarized EUV radiation toward the second mirror (e.g., the mirror **132**), where the multilayer mirror (e.g., the multilayer mirror **402a**) is below the multilayer polarizer **502**.

(114) Although FIG. **13** shows example blocks of process **1300**, in some implementations, process **1300** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **13**. Additionally, or alternatively, two or more of the blocks of process **1300** may be performed in parallel.

(115) FIG. **14** is a flowchart of an example process **1400** associated with configuring a pixel of an illumination system described herein. In some implementations, one or more process blocks of FIG. **14** may be performed by an illumination system (e.g., the illumination system **130b**, the device **1100**). In some implementations, one or more process blocks of FIG. **14** may be performed by another device or a group of devices separate from or including the illumination system, such as a lithography system (e.g., the lithography system **100**), a radiation source (e.g., the radiation source **102**), an exposure tool (e.g., the exposure tool **104**), a controller (e.g., the controller **140**), and/or an actuator (e.g., an actuator **724a-724c**), among other examples. Additionally, or alternatively, one or more process blocks of FIG. **14** may be performed by one or more components of device **1100**, such as processor **1120**, memory **1130**, input component **1140**, output component **1150**, and/or communication component **1160**.

(116) As further shown in FIG. **14**, process **1400** may include forming a photosensitive material on a semiconductor substrate (block **1410**). For example, a deposition tool such as a spin-coating tool and/or another type of deposition tool may form a photosensitive material on a semiconductor substrate **110**, as described herein. The photosensitive material may include a photoresist, such as a positive photoresist, a negative photoresist, and/or another type of photoresist.

(117) As shown in FIG. **14**, process **1400** may include loading the semiconductor substrate after forming the photosensitive material on the semiconductor substrate (block **1420**). For example, the lithography system **100** may load the semiconductor substrate **110** onto a substrate stage **136** of the lithography system **100** after the deposition tool forms the photosensitive material on the semiconductor substrate **110**, as described herein.

(118) As further shown in FIG. 14, process **1400** may include configuring a plurality of multilayer mirrors and a multilayer polarizer of a configurable pixel included on a substrate of an illumination system (block **1430**). For example, the illumination system **130b** (e.g., the actuators **724a-724c** and/or **908** of the illumination system **130b**) may configure the multilayer mirrors **402a** and **402b** and the multilayer polarizer **502** of a configurable pixel **702** included on the substrate **206** of the illumination system **130b**, as described above. In some implementations, the illumination system **130b** is included in an EUV exposure tool (e.g., the exposure tool **104**) of the lithography system **100**. In some implementations, the plurality of multilayer mirrors **402a** and **402b** and the multilayer polarizer **502** are configured such that the configurable pixel **702** is configured in a particular polarization configuration (e.g., the polarization configurations described in connection with the example implementations **700**, **810-840**, and/or **900**).

(119) As further shown in FIG. 14, process **1400** may include modulating unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form polarized radiation from the unpolarized radiation (block **1440**). For example, the lithography system **100** may modulate unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form polarized radiation from the unpolarized radiation. In some implementations, the lithography system **100** modulates the unpolarized radiation by generating a laser beam **120**, which is used to generate EUV radiation **106** from droplets **118**, and by modulating the EUV radiation **106** that is provided to an exposure tool of the lithography system **100** where the plurality of multilayer mirrors and the multilayer polarizer are included.

(120) As further shown in FIG. 14, process **1400** may include exposing the photosensitive material on the semiconductor substrate to the polarized radiation (block **1450**). For example, the lithography system **100** may expose the photosensitive material on the semiconductor substrate **110** to the polarized radiation, as described herein. In some implementations, the polarized radiation is based on a particular polarization configuration of the configurable pixel.

(121) Process **1400** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(122) In a first implementation, the particular polarization configuration includes a transverse electric (TE) polarization configuration, and configuring the plurality of multilayer mirrors and the multilayer polarizer includes actuating a multilayer mirror, of the plurality of multilayer mirrors, below the multilayer polarizer such that the polarized radiation includes a TE polarized component of the unpolarized radiation.

(123) In a second implementation, alone or in combination with the first implementation, modulating the unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form the polarized radiation from the unpolarized radiation includes polarizing, using the multilayer polarizer, the unpolarized radiation into the TE polarized component and a transverse magnetic (TM) polarized component, and reflecting, using another multilayer mirror of the plurality of multilayer mirrors, the TE polarized component.

(124) In a third implementation, alone or in combination with one or more of the first and second implementations, the particular polarization configuration includes a transverse magnetic (TM) polarization configuration, and configuring the plurality of multilayer mirrors and the multilayer polarizer includes actuating a multilayer mirror, of the plurality of multilayer mirrors, adjacent to the multilayer polarizer such that the polarized radiation includes a TM polarized component of the unpolarized radiation.

(125) In a fourth implementation, alone or in combination with one or more of the first through third implementations, modulating the unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form the polarized radiation from the unpolarized radiation includes polarizing, using the multilayer polarizer, the unpolarized radiation into the TM polarized component and a transverse electric (TE) polarized component.

- (126) In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, the particular polarization configuration includes an unpolarized polarization configuration, and configuring the plurality of multilayer mirrors and the multilayer polarizer includes actuating the multilayer polarizer such that the unpolarized radiation is directed toward the semiconductor substrate.
- (127) In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, the multilayer mirror is below the multilayer polarizer.
- (128) Although FIG. **14** shows example blocks of process **1400**, in some implementations, process **1400** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **14**. Additionally, or alternatively, two or more of the blocks of process **1400** may be performed in parallel.
- (129) In this way, an illumination system includes a plurality of pixels (or spots) that are (or may be) configured in one or more polarization configuration types. The pixels of the illumination system may be configured to promote particular types of polarization (e.g., transverse electric (TE) polarization, transverse magnetic (TM) polarization) to increase pattern contrast while achieving suitable exposure operation throughput. Moreover, the pixels of the pixels of the illumination system may be configured to achieve free-form (arbitrary or freely-configurable) polarization, which permits the polarization of radiation to be tailored to particular exposure operation patterns and other parameters.
- (130) As described in greater detail above, some implementations described herein provide a method. The method includes forming a photosensitive material on a semiconductor substrate. The method includes loading the semiconductor substrate after forming the photosensitive material on the semiconductor substrate. The method includes configuring a plurality of multilayer mirrors and a multilayer polarizer of a configurable pixel included on a substrate of an illumination system. The method includes modulating unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form polarized radiation from the unpolarized radiation. The method includes exposing the photosensitive material on the semiconductor substrate to the polarized radiation. The polarized radiation is based on a particular polarization configuration of the configurable pixel.
- (131) As described in greater detail above, some implementations described herein provide a method. The method includes transmitting, by a controller, a first signal to cause at least one of a plurality of configurable pixels of an illumination system to be configured in a polarization configuration of a plurality of polarization configurations. The method includes transmitting, by the controller, a second signal to cause a lithography system to perform an exposure operation while the at least one of the plurality of configurable pixels is configured in the polarization configuration, where the illumination system is included in an exposure tool of the lithography system.
- (132) As described in greater detail above, some implementations described herein provide a method. The method includes configuring a plurality of multilayer mirrors and a multilayer polarizer of a configurable pixel included on a substrate of an illumination system, where the illumination system is included in an EUV exposure tool of a lithography system, and where the plurality of multilayer mirrors and the multilayer polarizer are configured such that the configurable pixel is configured in a particular polarization configuration. The method includes receiving unpolarized EUV radiation from a first mirror included in the EUV exposure tool during an exposure operation of the EUV exposure tool. The method includes providing reflected EUV radiation to a second mirror included in the EUV exposure tool during the exposure operation, where a polarization of the reflected EUV radiation is based on the unpolarized EUV radiation and the particular polarization configuration of the configurable pixel.
- (133) As described in greater detail above, some implementations described herein provide an illumination system for use in an EUV exposure tool. The illumination system for use in an EUV

exposure tool includes a substrate. The illumination system for use in an EUV exposure tool includes a plurality of fixed IL polarized pixels. The illumination system for use in an EUV exposure tool includes a plurality of configurable pixels that are each capable of being selectively configured in a respective polarization configuration of a plurality of polarization configurations, where the plurality of polarization configurations include a TE polarized configuration, a TM polarized configuration, a TE polarized and TM polarized configuration, or an unpolarized configuration.

(134) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a photosensitive material on a semiconductor substrate; loading the semiconductor substrate after forming the photosensitive material on the semiconductor substrate; configuring a plurality of multilayer mirrors and a multilayer polarizer of a configurable pixel included on a substrate of an illumination system, wherein the configurable pixel is capable of switching between different polarization configurations of a plurality of polarization configurations; modulating unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form polarized radiation from the unpolarized radiation; and exposing the photosensitive material on the semiconductor substrate to the polarized radiation, wherein the polarized radiation is based on a particular polarization configuration of the plurality of polarization configurations.
2. The method of claim 1, wherein the particular polarization configuration comprises a transverse electric (TE) polarization configuration; and wherein configuring the plurality of multilayer mirrors and the multilayer polarizer comprises: actuating a multilayer mirror, of the plurality of multilayer mirrors, below the multilayer polarizer such that the polarized radiation comprises a TE polarized component of the unpolarized radiation.
3. The method of claim 2, wherein modulating the unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form the polarized radiation from the unpolarized radiation comprises: polarizing, using the multilayer polarizer, the unpolarized radiation into the TE polarized component and a transverse magnetic (TM) polarized component; and reflecting, using another multilayer mirror of the plurality of multilayer mirrors, the TE polarized component.
4. The method of claim 2, wherein the particular polarization configuration comprises a transverse magnetic (TM) polarization configuration; and wherein configuring the plurality of multilayer mirrors and the multilayer polarizer comprises: actuating a multilayer mirror, of the plurality of multilayer mirrors, adjacent to the multilayer polarizer such that the polarized radiation comprises a TM polarized component of the unpolarized radiation.
5. The method of claim 4, wherein modulating the unpolarized radiation and using the plurality of multilayer mirrors and the multilayer polarizer to form the polarized radiation from the unpolarized radiation comprises: polarizing, using the multilayer polarizer, the unpolarized radiation into the TM polarized component and a transverse electric (TE) polarized component.
6. The method of claim 1, wherein the particular polarization configuration comprises an unpolarized polarization configuration; and wherein configuring the plurality of multilayer mirrors and the multilayer polarizer comprises: actuating the multilayer polarizer such that the unpolarized

radiation is directed toward the semiconductor substrate.

7. The method of claim 6, wherein the multilayer mirror is below the multilayer polarizer.

8. An illumination system for use in an extreme ultraviolet (EUV) exposure tool, comprising: a substrate; a plurality of fixed transverse electric (TE) polarized pixels; and a plurality of configurable pixels that are each capable of being selectively configured in a respective polarization configuration of a plurality of polarization configurations, wherein the plurality of polarization configurations comprise: a TE polarized configuration, a transverse magnetic (TM) polarized configuration, a TE polarized and TM polarized configuration, or an unpolarized configuration, and wherein one or more of the plurality of configurable pixels are capable of selectively switching between different polarization configurations of the plurality of polarization configurations.

9. The illumination system of claim 8, further comprising: a controller configured to selectively configure the plurality of configurable pixels in the respective polarization configuration.

10. The illumination system of claim 9, wherein a configurable pixel of the plurality of configurable pixels includes an actuator configured to: communicate with the controller to receive a signal; and actuate, based on the signal, at least one of a multilayer polarizer or a multilayer mirror to selectively configure the configurable pixel in a polarization configuration of the plurality of polarization configurations.

11. The illumination system of claim 8, wherein a configurable pixel of the plurality of configurable pixels comprises: a multilayer polarizer; and a plurality of multilayer mirrors.

12. The illumination system of claim 11, wherein the multilayer polarizer, when configured in the TE polarized and TM polarized configuration, is configured to: transmit a TM polarized component, of unpolarized EUV radiation incident toward the configurable pixel, toward a first multilayer mirror of the plurality of multilayer mirrors; and reflect a TE polarized component of the unpolarized EUV radiation toward a second multilayer mirror of the plurality of multilayer mirrors.

13. The illumination system of claim 11, wherein the configurable pixel further comprises: a tunable mirror configured to reflect EUV radiation, that is reflected from the multilayer polarizer, toward a multilayer mirror of the plurality of multilayer mirrors.

14. A method, comprising: transmitting, by a controller, a first signal to cause at least one of a plurality of configurable pixels of an illumination system to be configured in a polarization configuration of a plurality of polarization configurations, wherein the at least one of the plurality of configurable pixels is capable of switching between different polarization configurations of the plurality of polarization configurations; and transmitting, by the controller, a second signal to cause a lithography system to perform an exposure operation while the at least one of the plurality of configurable pixels is configured in the polarization configuration, wherein the illumination system is included in an exposure tool of the lithography system.

15. The method of claim 14, wherein transmitting the first signal comprises: transmitting the first signal to cause each of the plurality of configurable pixels to be configured in respective polarization configurations, of the plurality of polarization configurations, prior to the exposure operation.

16. The method of claim 14, wherein transmitting the first signal comprises: transmitting the first signal to cause each of the plurality of configurable pixels to be configured in respective polarization configurations, of the plurality of polarization configurations, during the exposure operation.

17. The method of claim 14, further comprising: determining a particular combination of respective polarization configurations, of the plurality of polarization configurations, for the plurality of configurable pixels based on at least one of: a contrast parameter for the exposure operation, or a throughput parameter for the exposure operation.

18. The method of claim 14, wherein the polarization configuration is a transverse electric (TE) polarized configuration.

19. The method of claim 14, wherein transmitting the first signal comprises: transmitting the first signal to cause: a first subset of the plurality of configurable pixels to be configured in a transverse electric (TE) polarized configuration of the plurality of polarization configurations, a second subset of the plurality of configurable pixels to be configured in a transverse magnetic (TM) polarized configuration of the plurality of polarization configurations, and a third subset of the plurality of configurable pixels to be configured in an unpolarized configuration.

20. The method of claim 14, wherein transmitting the first signal comprises: transmitting the first signal to respective one or more actuators associated with each of the plurality of configurable pixels to cause the respective one or more actuators to configure the plurality of configurable pixels in respective polarization configurations.
